

The background is a blurred image of a computer monitor. It shows a dark-themed code editor with various lines of code in different colors (blue, green, white). At the bottom, a taskbar is visible with several application icons, including what looks like a web browser, a file explorer, and some development tools. The overall image is out of focus, serving as a backdrop for the text.

C#

Level 1

if (if then) Statement



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if (if then) Statement

Definition

if Statement

-  Executes a block of code only when a condition is true.

-  Used for decision making in *C#*.

Explanation

if

-  Tests one condition.
-  If the condition is true, the code inside the block executes.
-  If the condition is false, the block is skipped.

if...else

-  Provides two possible outcomes.
-  If the condition is true, the if block executes.
-  If the condition is false, the else block executes.

if...else if...else

-  Used to test multiple conditions.
-  Conditions are checked from top to bottom.
-  The first true condition is executed, then the remaining conditions are skipped.

Logical Operators

-  && requires all conditions to be true.
-  || requires at least one condition to be true.

Procedures

- 1 ☐ Read values for x and y.
- 2 ☐ Test whether both x and y are zero using if.
- 3 ☐ Test the same condition using if...else.
- 4 ☐ Test multiple possibilities using if...else if...else.
- 5 ☐ Execute the matching block.

💡 Example

```
if (x == 0 && y == 0)
{
    Console.WriteLine("Both x and y are zero");
}
else if (x == 0)
{
    Console.WriteLine("x is zero");
}
else if (y == 0)
{
    Console.WriteLine("y is zero");
}
else
{
    Console.WriteLine("Neither x nor y is zero");
}
```

➡ Result

Enter a number X:

55

Enter a number Y:

0

y is zero

Note

- ◆ Conditions in an if statement must produce a Boolean value: true or false.
- ◆ Multiple conditions can be combined using logical operators.
- ◆ In an if...else if...else chain, only the first true condition is executed.



Comparison



if

✓ Executes code only when the condition is true.



if...else

✓ Chooses between two possible outcomes.



if...else if...else

✓ Chooses between multiple possible outcomes.

✦ Summary

- ✓ if tests a condition and executes code when it is true.
 - ✓ else handles the false case.
 - ✓ else if allows testing additional conditions.
 - ✓ && means all conditions must be true.
 - ✓ || means at least one condition must be true.
- logical operators.

Thank you 😊